# DSGE Model Estimation via VAR IRF Matching: The Short and Long of It

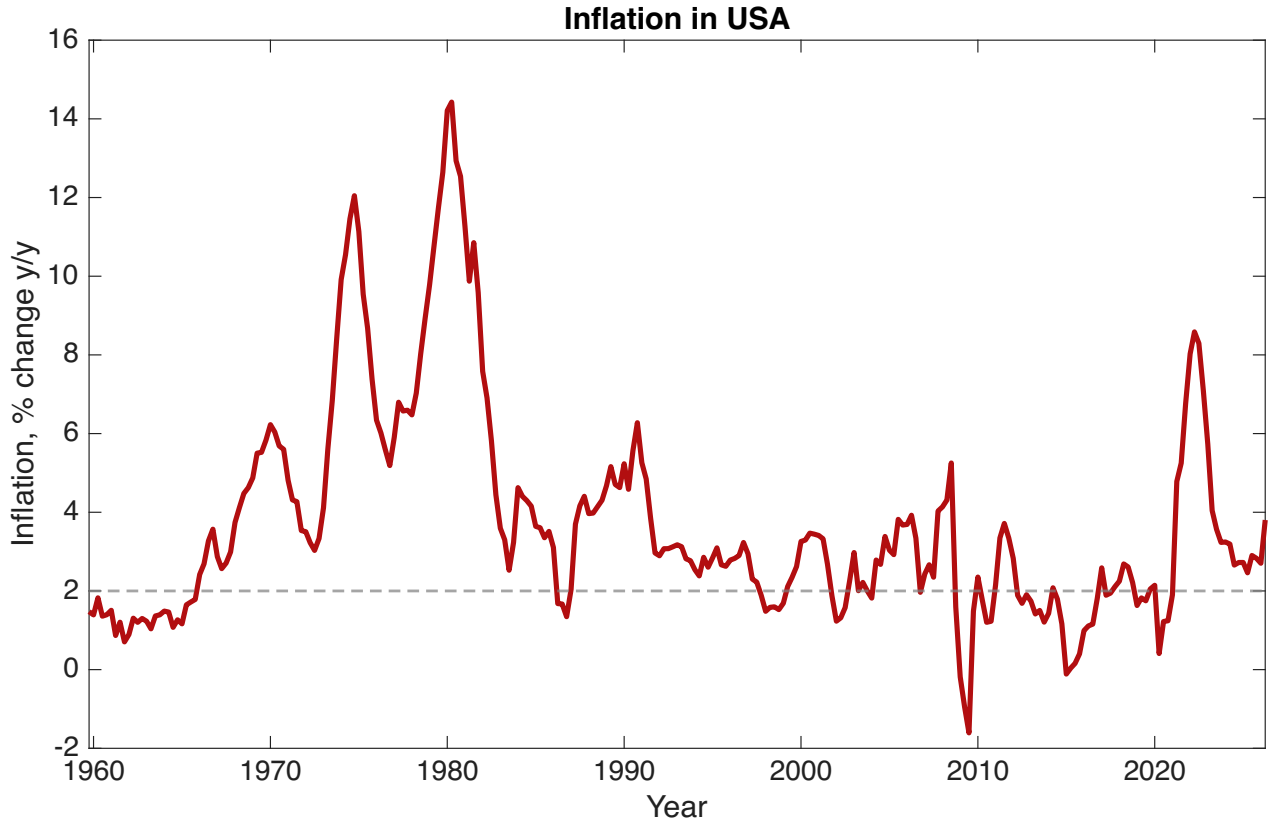
Lorenzo Del Col

Supervisor: Professor Efrem Castelnuovo

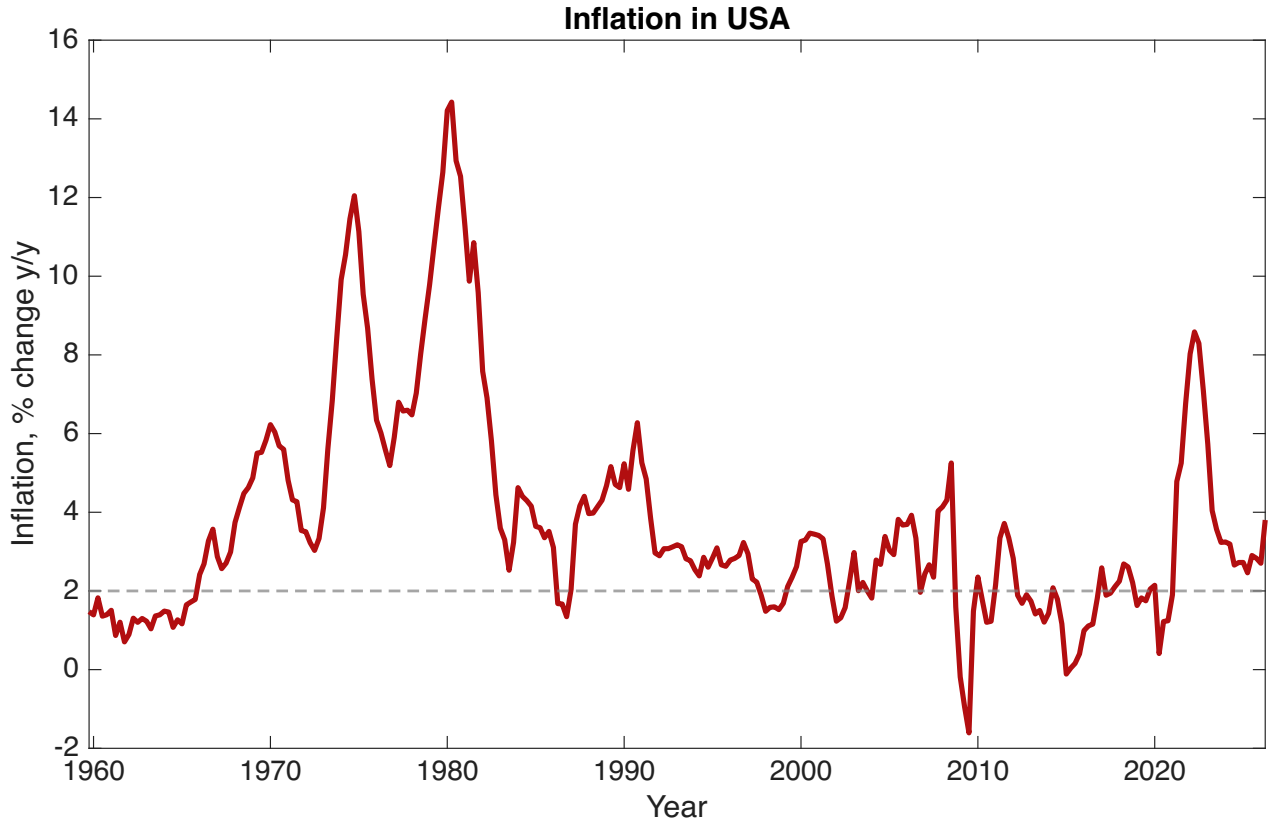
*University of Padova*
*MSc Applied Economics*

October 14, 2026

# Stylised Fact: Inflation Persistence

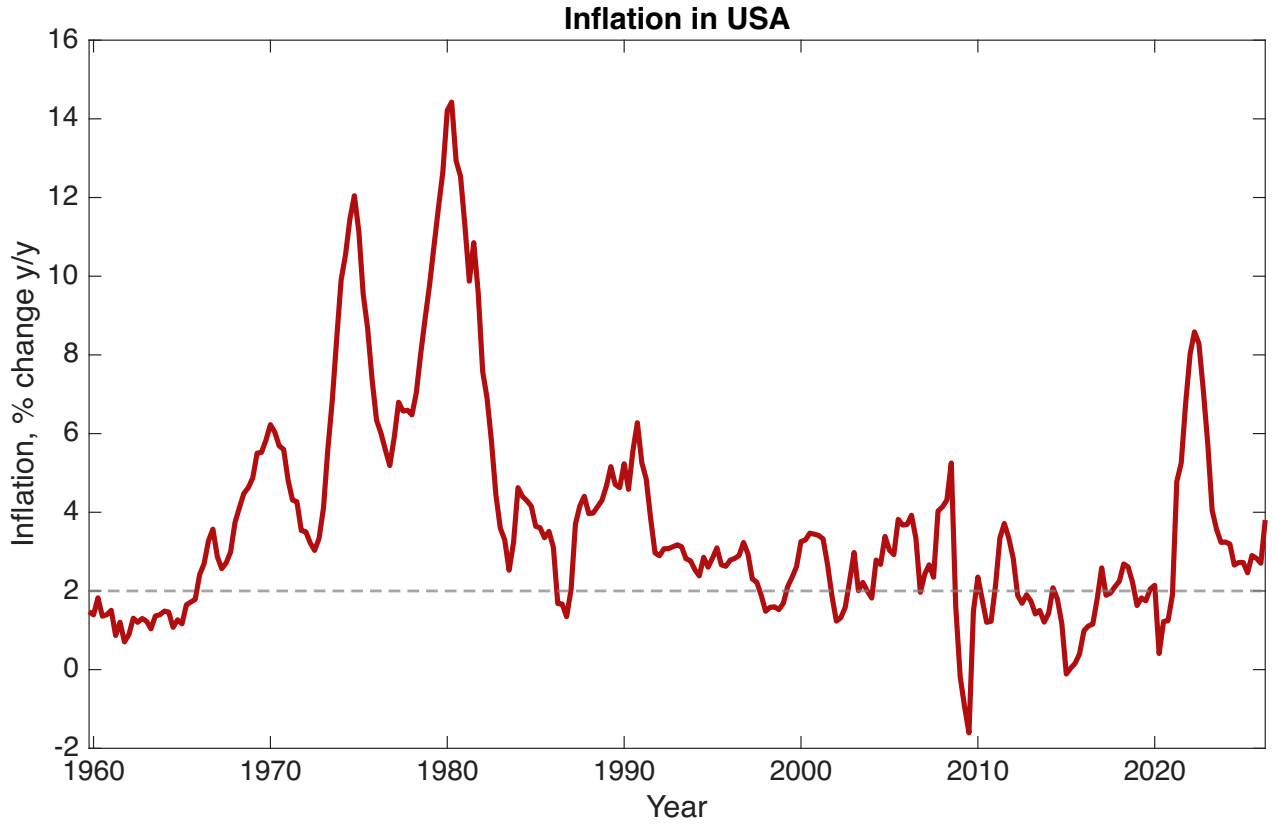


# Stylised Fact: Inflation Persistence




- DSGE models are often used to explain inflation persistence

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- DSGE models are often used to explain inflation persistence
- But standard NK models reproduce it with **unrealistically high levels of price stickiness**, inconsistent with micro data evidence

# CEE and ACEL's DSGE Motivation

- CEE JPE (2005) and ACEL RoED (2011) seminal works introduced the concept of **Firm Specific Capital** to model inflation persistence, mechanism supported by IRFs matching with VAR(4)

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- **Impulse Response Function Matching Estimation (IRFME)** is a common way to estimate DSGE models through VARs

$$\hat{\zeta} = \arg\min_{\zeta} (\hat{\Psi} - \Psi(\zeta))' V^{-1} (\hat{\Psi} - \Psi(\zeta)) \tag{1}$$

# Motivation

- *"If the* ***VAR is misspecified****, then, the* ***specification errors will be computed at each horizon****"*

**Ramey (2016) p. 83**

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- *"If the* ***VAR is misspecified****, then, the* ***specification errors will be computed at each horizon****"*

  **Ramey (2016) p. 83**

- **Truncation Bias is a type of misspecification** that arises when the VAR lag order is shorter than its DGP.

- Misspecification can deliver distorted results misinforming the central banker

# This Paper

**Research Question:**

- Can VAR **truncation bias** have significant implications for DSGE estimation?

**What's been done:**

- Revisit ACEL contribution, estimating Homogeneous vs Firm Specific Capital versions of their DSGE with VARs with different dynamic structures

# This Paper

**Research Question:**

- Can VAR **truncation bias** have significant implications for DSGE estimation?

**What's been done:**

- Revisit ACEL contribution, estimating Homogeneous vs Firm Specific Capital versions of their DSGE with VARs with different dynamic structures

**Findings:**

- Increasing the VAR lag structure casts doubts on the importance of Specificity of Capital to explain inflation inertia
- Lag choice can have implications for optimal policy design, modelling and structural inference

# Roadmap

- Related Literature
- ACEL model
- Truncation Bias
- Results

# Related Literature

**Price Stickiness:**

- Altig et al., (2011); Bils and Klenow, (2004); Golosov and Lucas, (2007); Klenow and Kryvtsov, (2008); Nakamura and Steinsson, (2008)

**Long-Lag VARs:**

- De Graeve and Westermark, (2025); Baumeister, (2025); Antolín-Díaz and Surico, (2025); Antolín-Díaz et al., (2026); Plagborg-Moller and Wolf, (2021); Ludwig, (2026)

**Truncation Bias and the DSGE-VAR mapping:**

- Ravenna, (2007); De Graeve and Westermark, (2025); Chari et al., (2008); Barnichon et al., (2026); Fernandez-Villaverde et al., (2007); Faust and Leeper, (1997); Lippi and Reichlin, (1993); Forni et al (2016)

**Impulse Response Matching and DSGE Estimation:**

- Rotemberg and Woodford, (1997); Christiano et al., (2005); Altig et al., (2011); Iacovello (2005); Bovin and Giannoni (2006), Hall et al (2012); Giacomini (2013); Theodoridis (2011); Guerron-Quintana (2010)

# ACEL's DSGE Mechanism

ACEL builds two models on Christiano et al. JPE, (2005):

- Homogeneous capital
- Firm-specific capital

- **Inflation is inertial** despite firms re-optimising prices on average every 1.5 quarters
- When firms re-optimise, **prices are adjusted by a small amount** because each firm's short run marginal cost is increasing in its own output
- The key assumption is that, in any given period, a **firm's capital stock is pre-determined and firm-specific**

# ACEL's DSGE Mechanism (Cont'd)

- When log-linearized, the only difference in the characterizing equilibrium equations lies in the Phillips Curve

$$\Delta\hat{\pi}_t = \mathbb{E}[\beta\Delta\hat{\pi}_{t+1} + \gamma\hat{s}_t|\Omega_t] \tag{2}$$

$$\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\chi \tag{3}$$

- **Models are observationally equivalent in terms of prices and quantities:** macro data identify $\gamma$, the difference between the two models lies in the implied $\xi_p$
- $\hat{s}_t$ is the economy-wide marginal cost, $\gamma$ the Phillips Curve slope, $\xi_p$ the Calvo price stickiness parameter
- $\chi$ reduces the sensitivity of inflation to marginal costs in the firm-specific capital model

# ACEL's DSGE Mechanism (Cont'd)

Here $\chi$ is a non-linear function representing the strategic complementarity in price settings

- In the **Homogeneous** model, $\chi = 1$: $\gamma$ depends only on $\beta$
- In the **Firm-specific** model, $\chi < 1$ and captures two forces that make a firm move its price *less* when it can reset:
  - It depends on Variable Capital Utilisation $\sigma_a$
  - The costlier it is to vary capital utilization, the steeper the firm's own marginal cost curve
  - The more elastic the firm's demand, the larger the fall in output (and hence in marginal cost) after a price increase

- So $\chi$ governs **how much** a firm moves its price
- $\xi_p$ **how often**, with implied duration $1/(1-\xi_p)$

Given a fixed value of $\gamma$, a smaller $\chi$ implies a smaller $\xi_p$

# ACEL's Monetary Policy Shock Propagation Mechanism

$\varepsilon_t^M \downarrow \Rightarrow i_t \downarrow \Rightarrow y_t^d \uparrow \Rightarrow y_t^s \uparrow \Rightarrow h_t^d \uparrow$
$\Rightarrow$ Calvo wages: $w_t \uparrow$, $\frac{w_t}{P_t} \uparrow$ little

**Homogeneous**

- $K_t = u_t \bar{K}_t \Rightarrow K_t^d \uparrow$ but $\hat{K}_t$ and $u_t \uparrow$ $r_t \uparrow$ given $\hat{\sigma}_a$
- $\Rightarrow MC_{i,t} = MC_t \Rightarrow$ MC independent from $y_{i,t}$ $\chi = 1$
- $MC_t \uparrow \Rightarrow P_t^{1-\xi_p} \uparrow \Rightarrow P_t \uparrow \Rightarrow \pi_t \uparrow$
- $\gamma = f(\bar{\xi}_p)$ given $\bar{\gamma}$ low, $\xi_p$ need to be high enough

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**Firm Specific**

- $K_{i,t} = u_{i,t}\bar{K}_{i,t} \uparrow \Rightarrow K_t^d \uparrow$ but $\hat{K}$
- Given $\chi < 1 \Rightarrow MC_{i,t} \uparrow$ in $y_{i,t}$
- $MC_{i,t} \uparrow \Rightarrow P_t^{1-\xi_p} \uparrow \Rightarrow P_t \uparrow \Rightarrow \pi_t \uparrow$
- $\gamma = f(\bar{\xi}_p; \bar{\sigma}_a)$ given $\bar{\gamma}$

- If $\sigma_a \to 0 \Rightarrow \chi \to 1 \Rightarrow \gamma = f(\bar{\xi}_p; 0) \Rightarrow \xi_{p,\mathrm{FS}} \to \xi_{p,\mathrm{H}}$

# ACEL (2005) Results

The estimated $\gamma$ is 0.040, which implies:

- **Homogeneous** capital: $\xi_p = 0.82$, firms change prices every 5.6 quarters
- **Firm-specific** capital: firms change prices every 1.5 quarters

**Only the firm-specific capital model reconciles the micro-macro pricing conflict**

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**Only the firm-specific capital model reconciles the micro-macro pricing conflict**

- **Are these results robust to an enriched lag structure of the VAR?**

# Illustration of the problem: Truncation Bias

$$Y_t = B_1 Y_{t-1} + \cdots + B_p Y_{t-p} + u_t, \qquad \mathbb{E}(u_t u_t') = \Sigma \tag{4}$$

$$B(L) = B_1 L + \cdots + B_p L^p \tag{5}$$

- The VAR with $p < p^*$ suffers from truncation bias.[1]
- Lag truncation biases the reduced form coefficients $B(L)$ and the reduced form variance covariance matrix $\Sigma$, which are the two main ingredients for IRF estimation.

$$\text{IRF} = f\big(B(L), \Sigma\big) \tag{6}$$

[1] Where $p^*$ denotes the DGP true lag length.

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**Lower Bias**

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**Lower Bias**

- By reducing the truncation bias misspecification at each horizon

**Lower Variance**

- Misspecification reduction lowers the variance of the residuals
- This variance reduction counteracts the imprecision caused by increased parametrisation

# Long-Lag VARs Benefits

The variance-reducing effect of lower misspecification often dominates the penalty of increased parameterisation.

Compared to short-lag VARs, Long-Lag VARs have:

- Reduced truncation bias
- More precise inference
- Lower MSE
- Better coverage

# Residuals' Variance Reduction

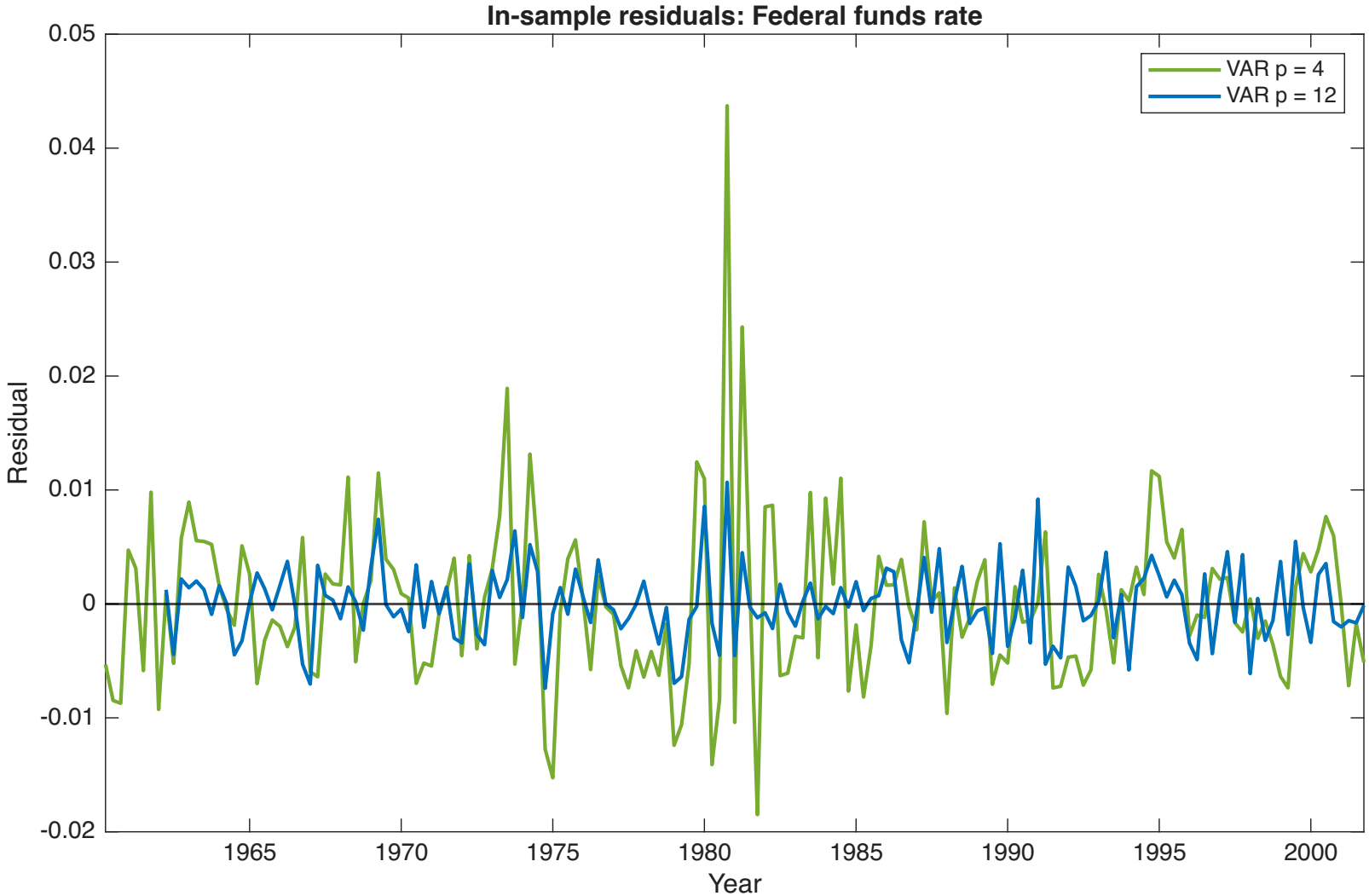
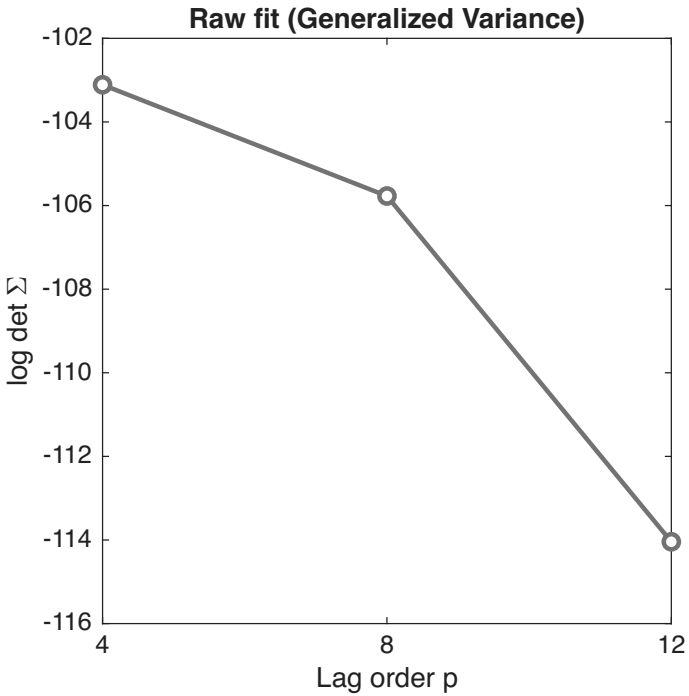
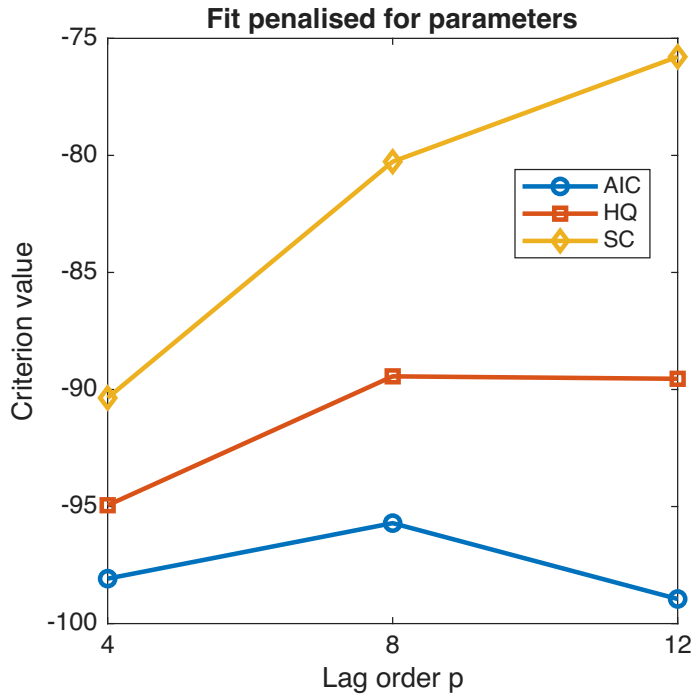



Figure 1: FFR Residuals Comparison Across Lags 4 vs 12

# Generalised Variance and Information Criteria




| **Lag** | $\log\det\hat{\Sigma}$ | **AIC** | **HQ** | **SC** |
|---|---|---|---|---|
| $p = 4$ | -103.1111 | -98.0796 | **-94.9444** | **-90.3591** |
| $p = 8$ | -105.7716 | -95.7087 | -89.4383 | -80.2677 |
| $p = 12$ | -114.0426 | **-98.9483** | -89.5426 | -75.7867 |

*Note:* Bold marks the lag order each criterion selects (lowest value). Selected lag: AIC $\rightarrow p = 12$, HQ $\rightarrow p = 4$, SC $\rightarrow p = 4$.

# Replication Exercise

Framework:

- **Altig et al., (2011)**'s DSGE[2]

VAR:

- Short and Long-Run restrictions, medium scale 10 variables
- Monetary Policy, Neutral and Embodied Technology shocks
- Sample period 1959:Q2-2001:Q4

Exercise:

- I re-estimate the VAR increasing lag order
- Re-estimate the DSGE by IRFs matching
- **Analyse implications of possible variations in the structural parameters and IRFs across different lag choices**

[2]Where the replication package provided by the authors refers to the 2005 NBER working paper

# VAR IRFs; Monetary Policy Shock

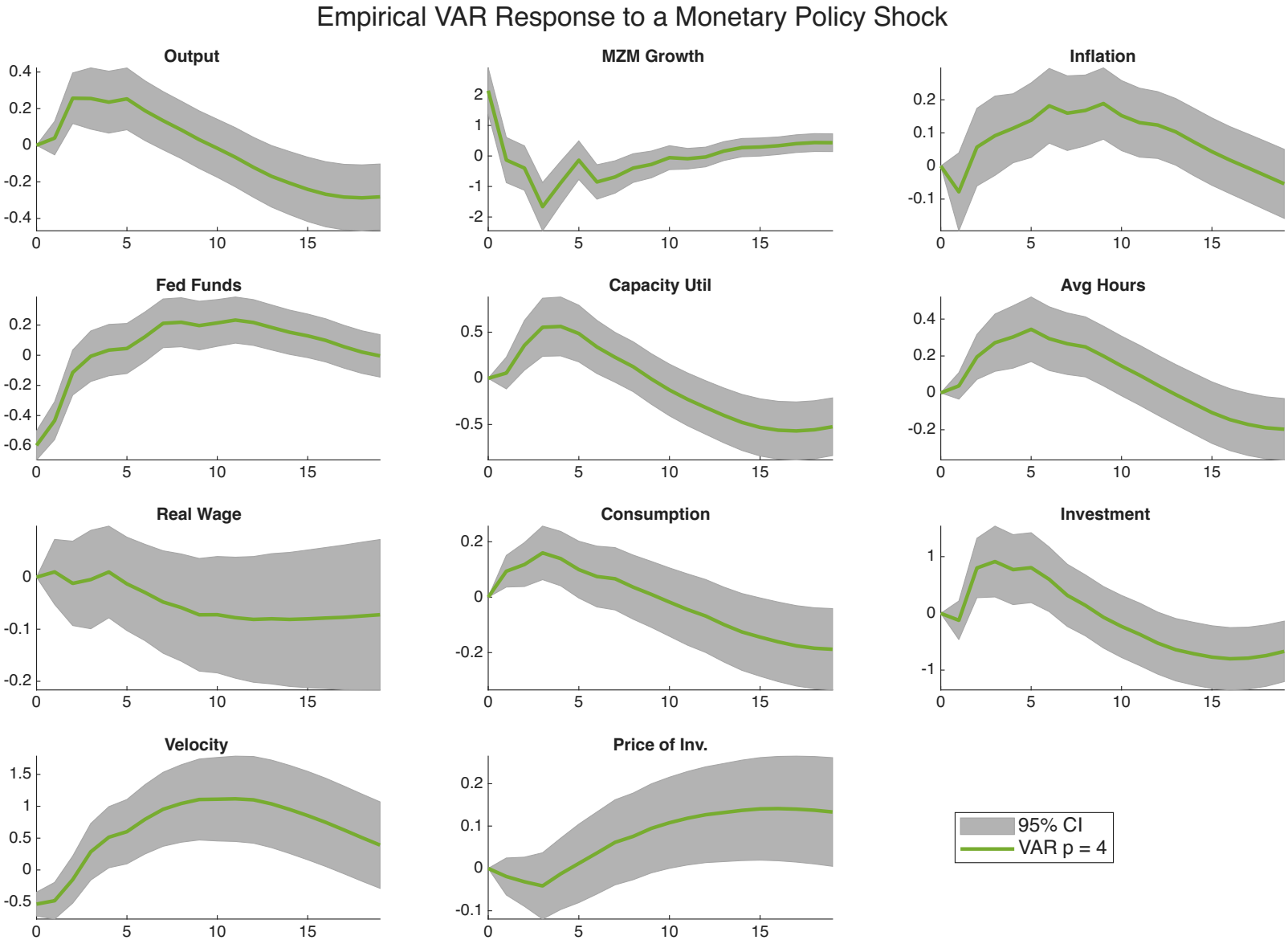



Figure 2: VAR IRFs Comparison: Monetary Policy Shock: 4 lags

# VAR IRFs; Monetary Policy Shock

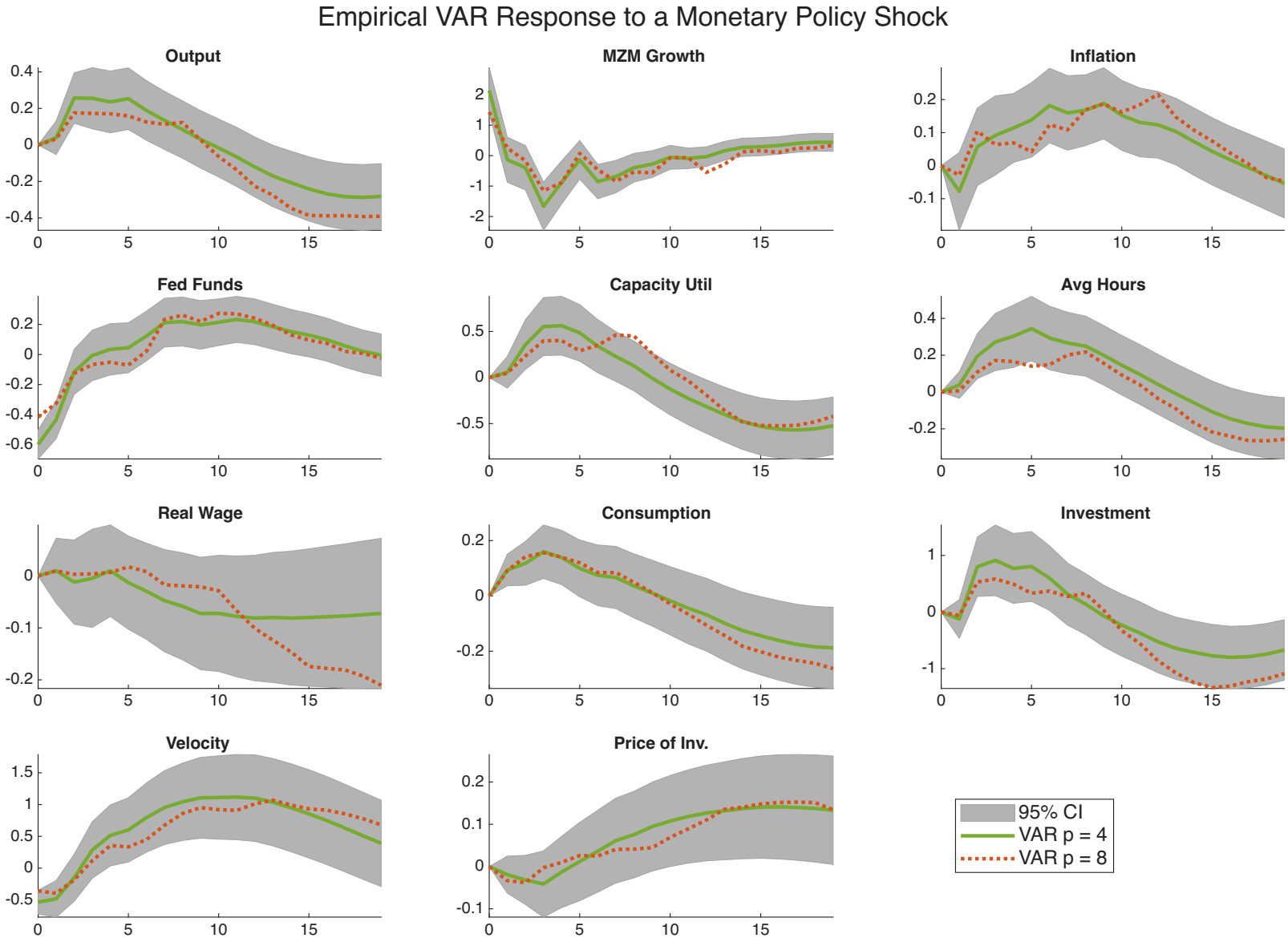



Figure 2: VAR IRFs Comparison: Monetary Policy Shock: 8 lags

# VAR IRFs; Monetary Policy Shock

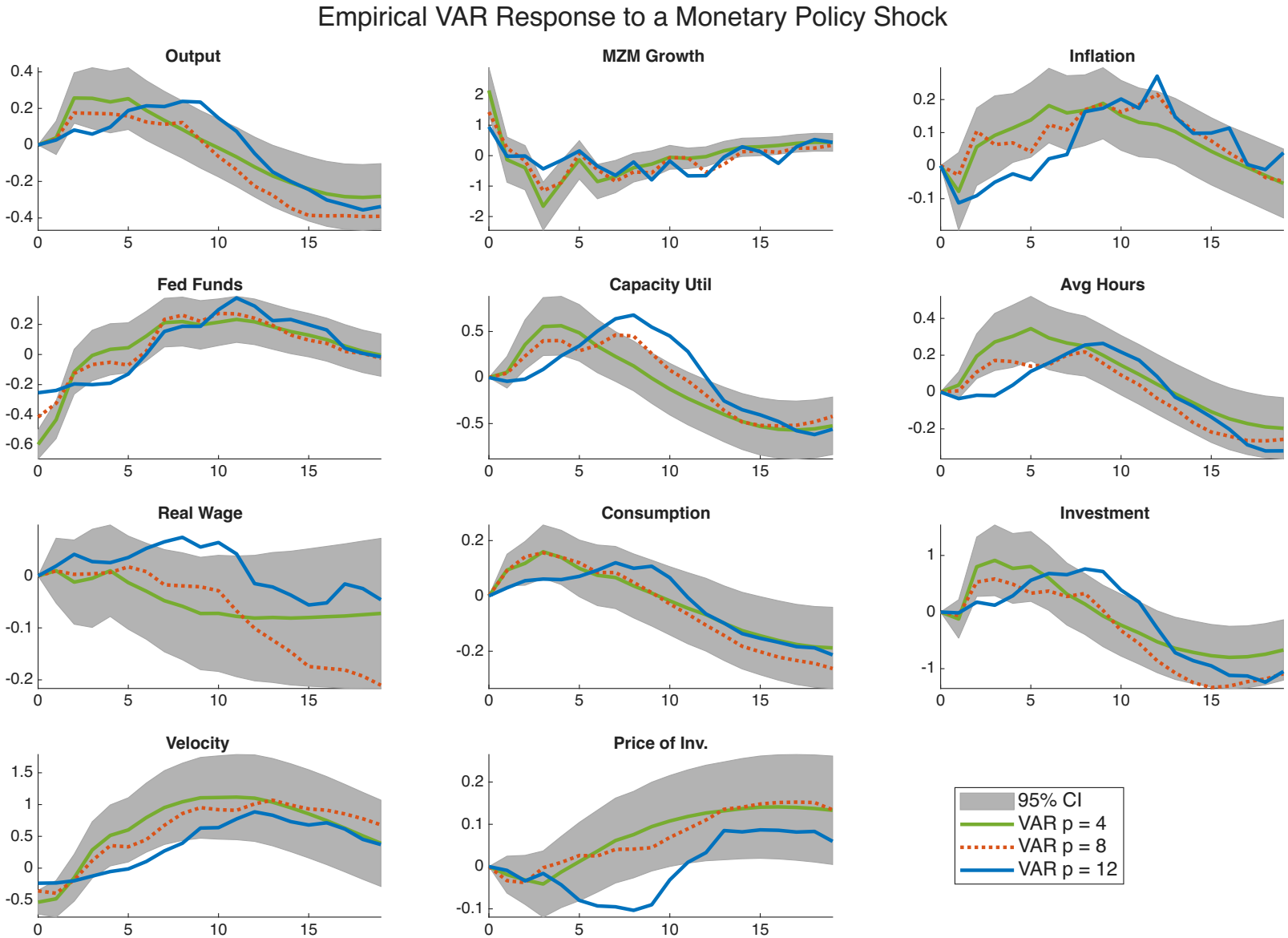



Figure 2: VAR IRFs Comparison: Monetary Policy Shock: 12 lags

# VAR IRFs; Monetary Policy Shock

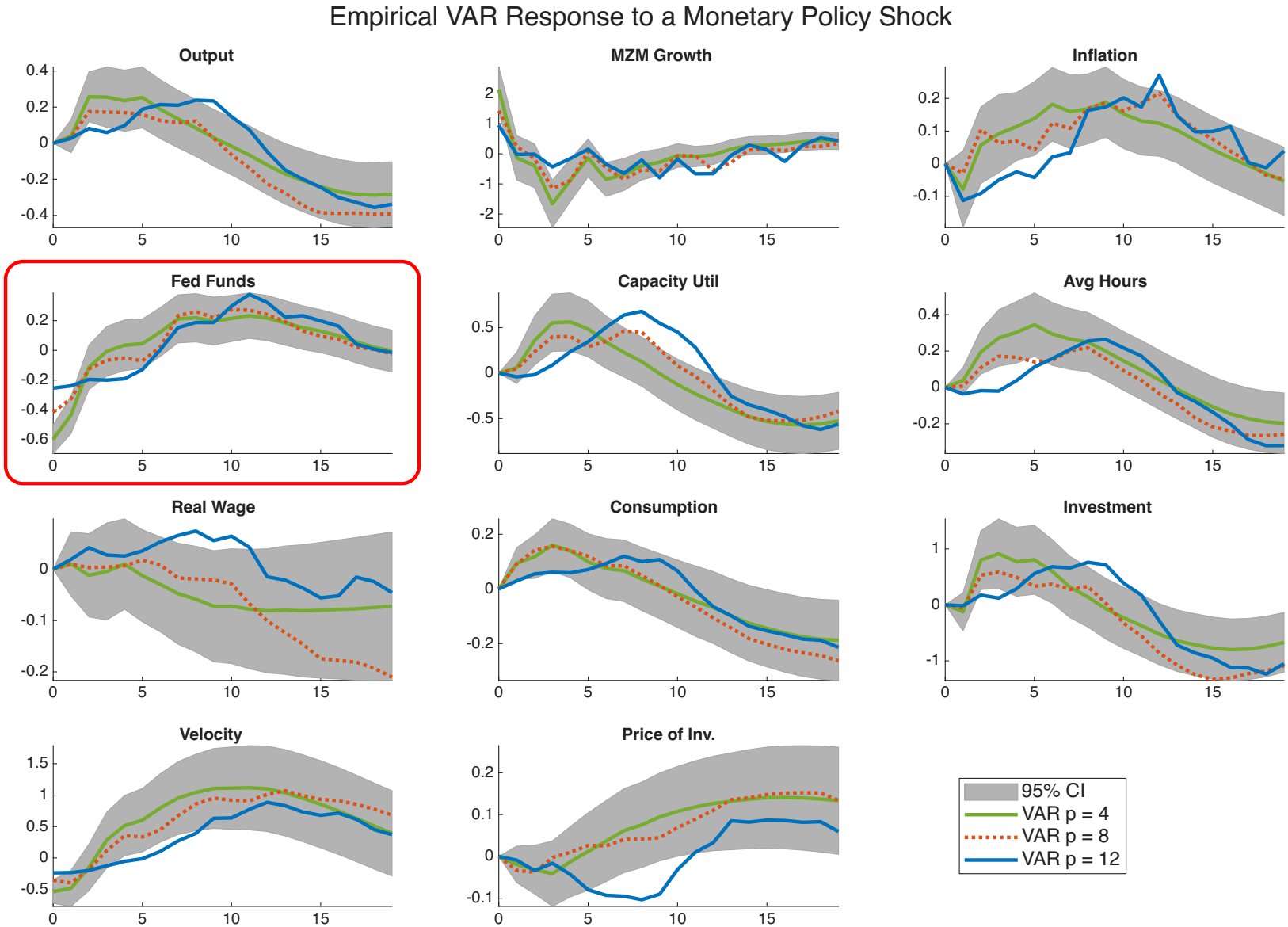



Figure 2: VAR IRFs Comparison: Monetary Policy Shock: 12 lags

# VAR IRFs; Monetary Policy Shock

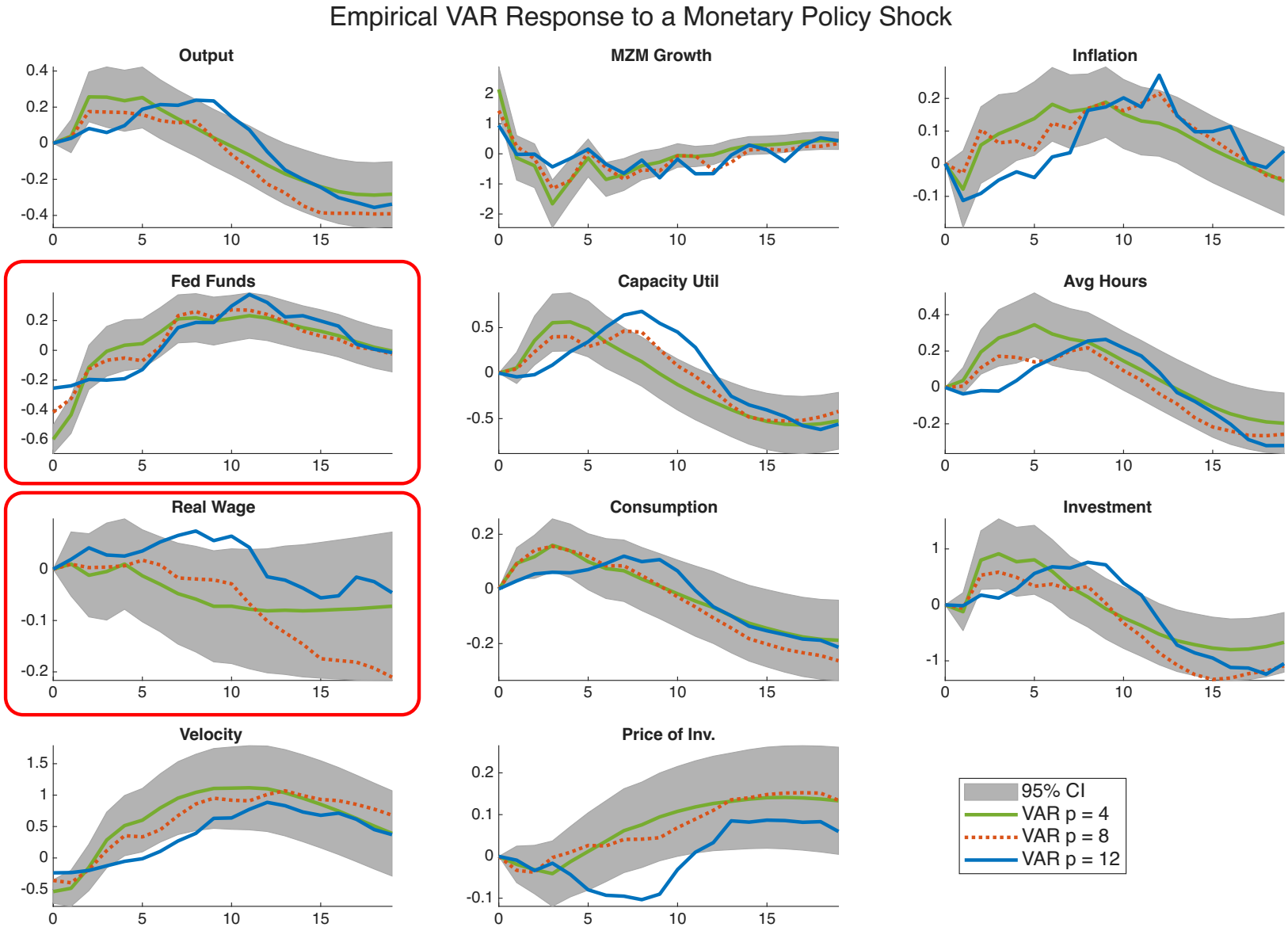



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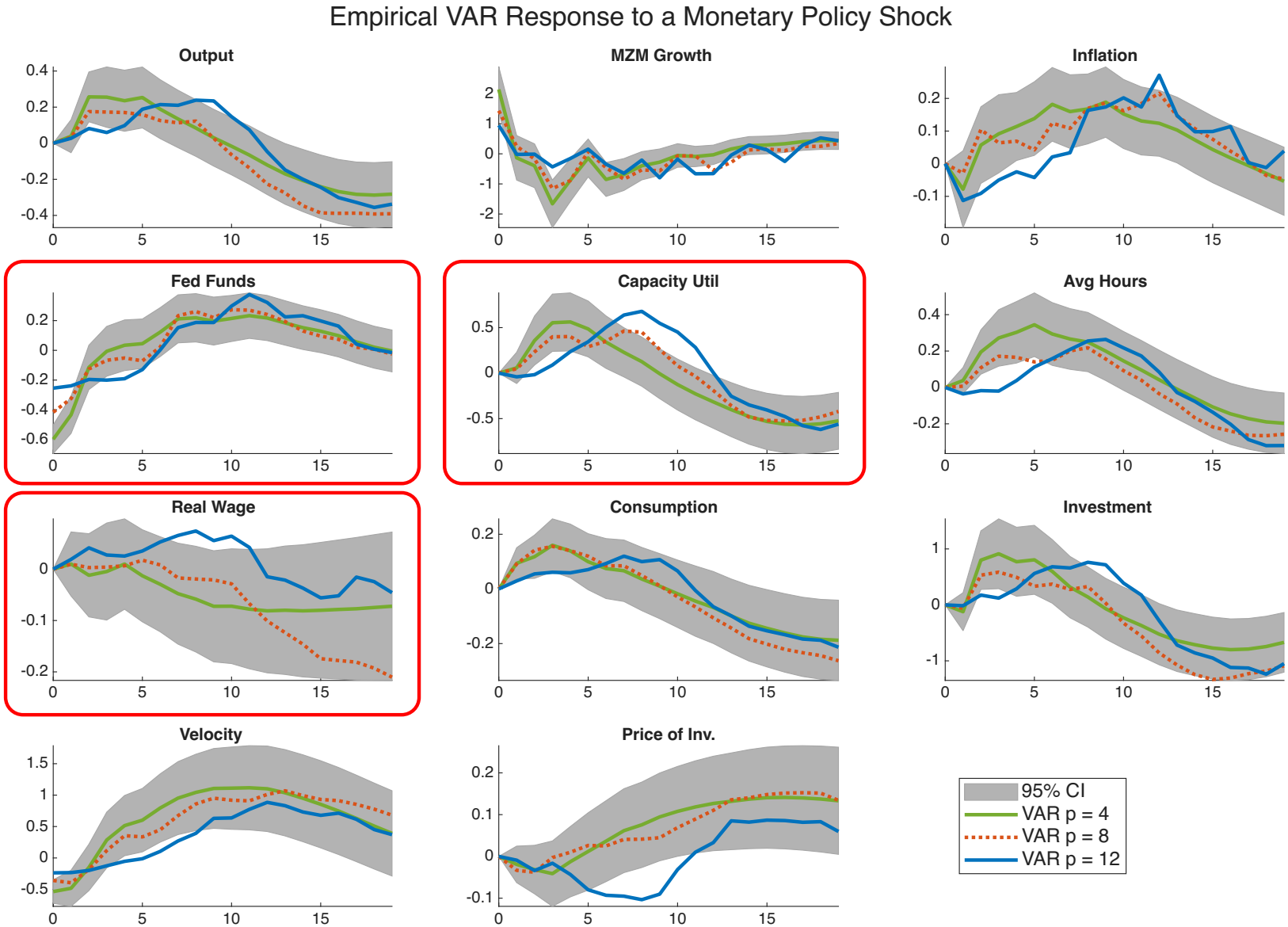



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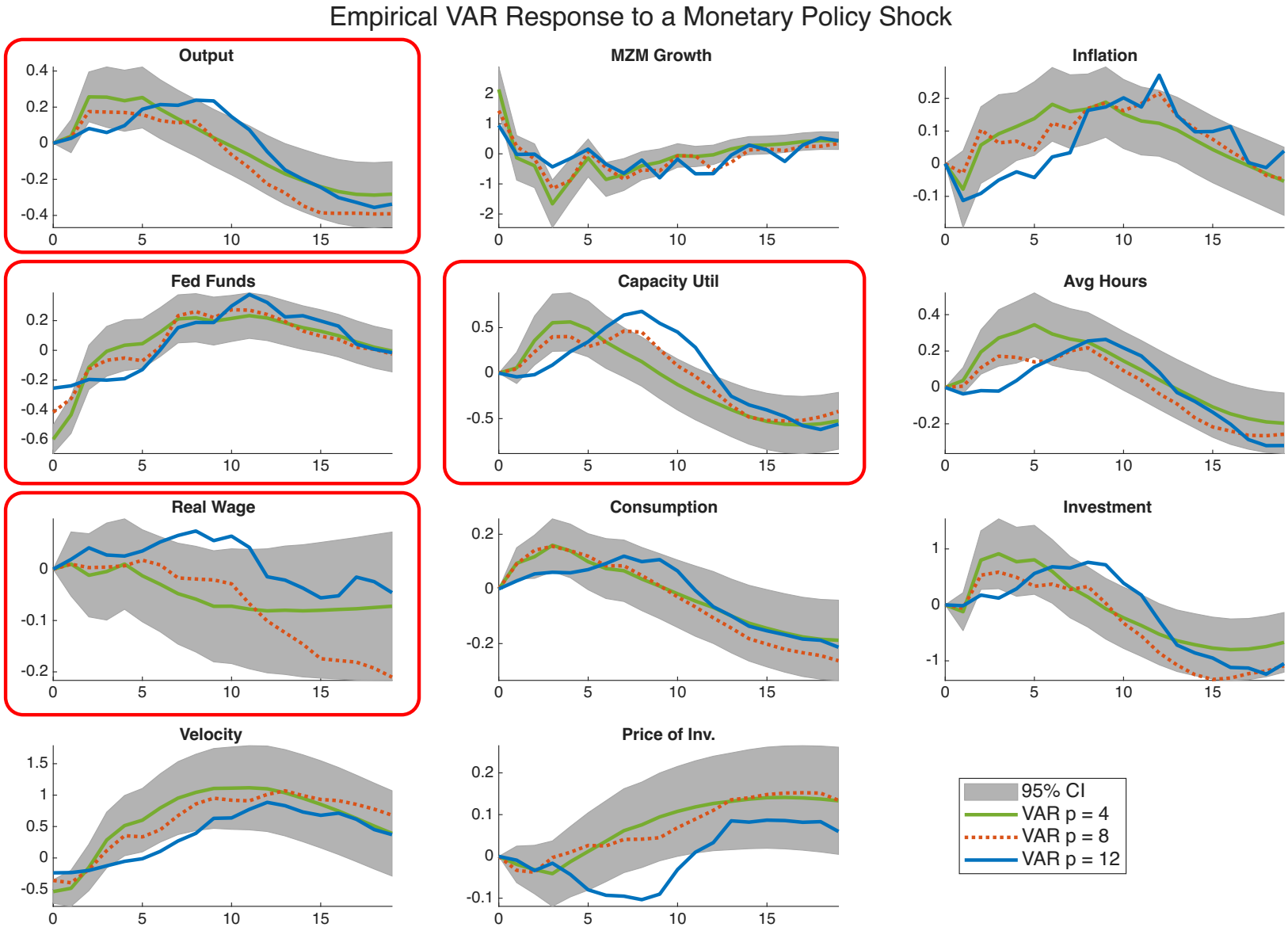



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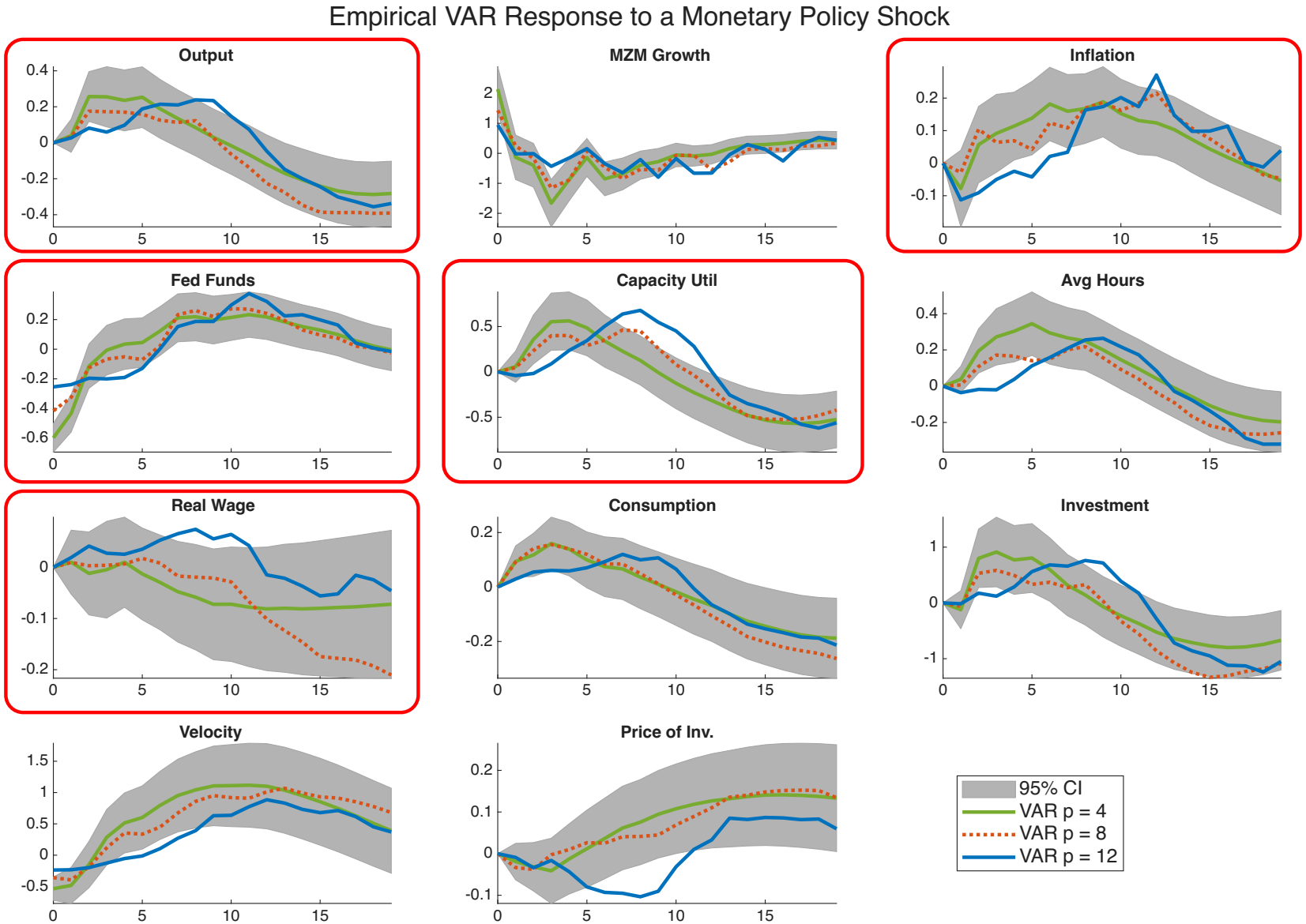



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# DSGE Structural Parameters

MonteCalvo

| **Parameter** | **VAR** $p = 4$ | **VAR** $p = 8$ | **VAR** $p = 12$ |
|---|---|---|---|
| *Structural Estimated Frictions* | | | |
| Price markup ($\lambda_f$) | 1.0100 | 1.0100 | 1.0100 |
| | (-) | (-) | (-) |
| Wage stickiness ($\xi_w$) | 0.7221*** | 0.5617*** | 0.4838*** |
| | (0.1506) | (0.1142) | (0.0757) |
| Phillips curve slope ($\gamma$) | 0.0402 | 0.0402 | 0.0129*** |
| | (0.0246) | (0.0598) | (0.0042) |
| Capacity util. cost ($\sigma_a$) | 1.9938 | 0.1175 | 0.0042 |
| | (2.6066) | (0.6299) | (0.0230) |
| Habit formation ($b$) | 0.7066*** | 0.6935*** | 0.8596*** |
| | (0.0463) | (0.0570) | (0.0203) |
| Inv. adj. cost ($S''$) | 3.2807* | 3.0372* | 6.8431*** |
| | (1.8874) | (1.5506) | (2.3306) |
| Money demand elast. ($\epsilon$) | 0.8077*** | 0.7531*** | 0.8346*** |
| | (0.2111) | (0.1473) | (0.1944) |
| *Implied Price Stickiness* | | | |
| Homogeneous capital ($\xi_{p,hom}$) | 0.8213*** | 0.8213*** | 0.8958*** |
| | (0.0480) | (0.0574) | (0.0139) |
| Firm-specific capital ($\xi_{p,fs}$) | 0.3372** | 0.6448*** | 0.8884*** |
| | (0.1505) | (0.1285) | (0.0220) |

*Note:* Standard errors in parentheses. The Calvo parameters are recovered numerically from the estimated structural parameters; their standard errors come from a 1,000,000-draw Monte Carlo of the joint distribution of the estimated structural parameters.

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# Implied Duration

| **Model Specification** | **VAR** $p = 4$ | **VAR** $p = 8$ | **VAR** $p = 12$ |
|---|---|---|---|
| *Homogeneous Capital Model* | | | |
| Calvo parameter ($\xi_{p,hom}$) | 0.82 | 0.82 | 0.90 |
| Implied duration (quarters) | **5.6** | **5.6** | **9.6** |
| *Firm-Specific Capital Model* | | | |
| Calvo parameter ($\xi_{p,fs}$) | 0.34 | 0.64 | 0.89 |
| Implied duration (quarters) | **1.5** | **2.8** | **9.0** |

*Note:* Implied duration is calculated as $D = 1/(1 - \xi_p)$.

- Firm Specific model's Calvo Price parameter converges to the Homogeneous one
- Back to the micro-macro implied price duration mismatch

$\chi$ Across Lags

# Conclusions

Following the advice from the Long-Lag VARs literature. Replicating ACEL et al., (2005), enriching the VAR lag structure I find that:

- Capacity Utilisation Cost, crucial variable to justify the firm specific model, dramatically shrinks to 0
- Firm specificity of capital seems not the key concept to explain inflation inertia
- ACEL's evidence on firm specific capital is not robust to long-lag VARs

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**De Graeve & Westermark (2025) and Baumeister (2025)**

- Long-Lag VARs might have structural implications, it's up to the researcher to demonstrate that their findings are robust to an enriched lag structure

**Thanks a lot for your attention!**

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# Implied Real Rigidity ($\chi$) Across Lags

| **Model** | **VAR** $p = 4$ | **VAR** $p = 8$ | **VAR** $p = 12$ |
|---|---|---|---|
| $\chi_{hom}$ $(\equiv 1)$ | 1.0000<br>(-) | 1.0000<br>(-) | 1.0000<br>(-) |
| $\chi_{fs}$ | 0.0307<br>(0.0596) | 0.2030<br>(0.1667) | 0.8671***<br>(0.1698) |

*Note:* Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. $\chi = \gamma\xi_p/[(1-\xi_p)(1-\beta\xi_p)]$, recovered by inverting the Phillips curve slope equation; $\chi_{hom} \equiv 1$ by construction. Standard errors for $\chi_{fs}$ come from the same 1,000,000-draw joint Monte Carlo used for the Calvo parameters.

Back to Implied Duration

# VAR IRFs All CI; Monetary Policy Shock

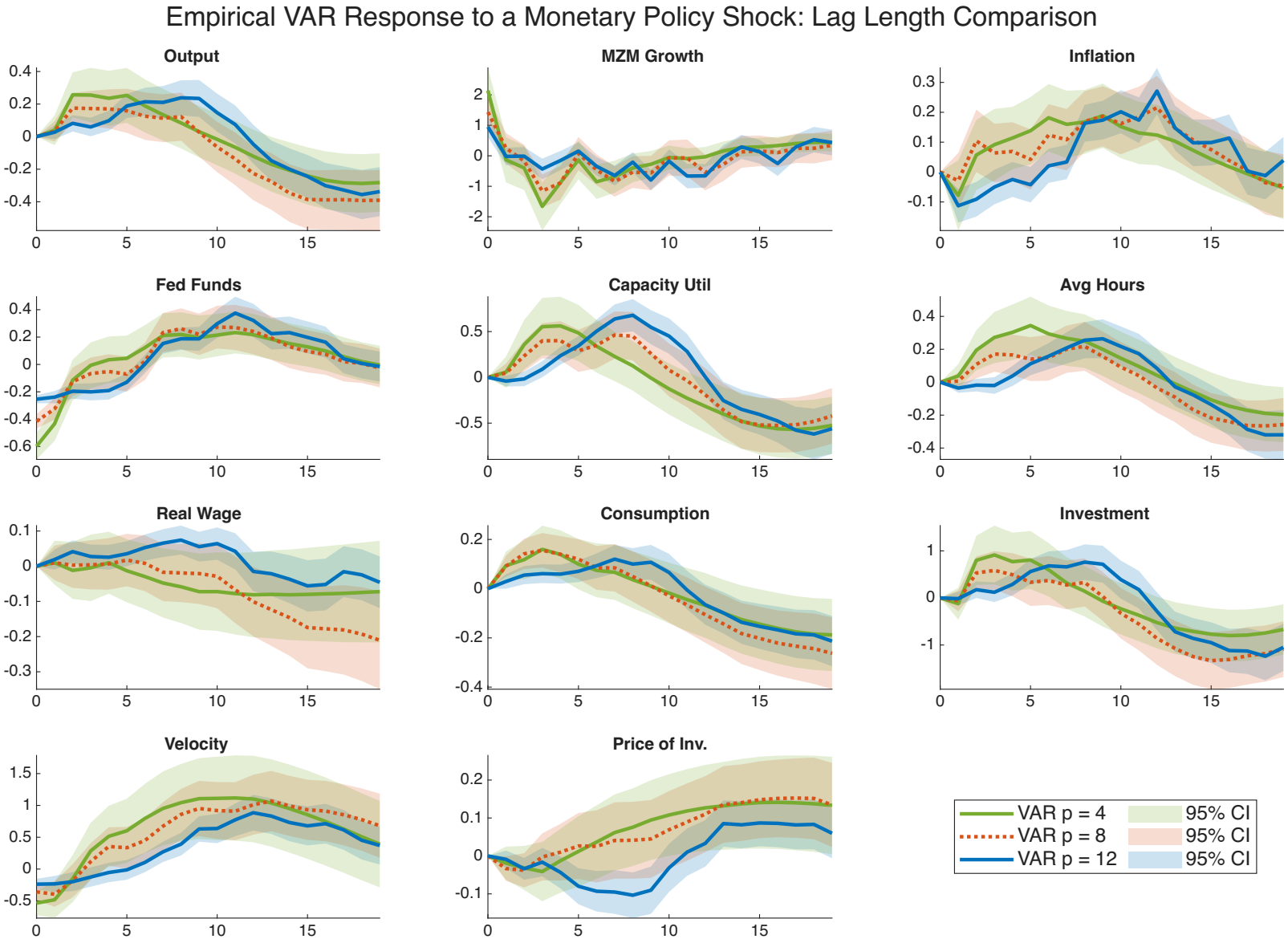



Figure 3: IRFs VARs all CI Comparison: Monetary Policy Shock

# Bootstrap Confidence Intervals Shrinks!

- With respect to De Graeve and Westermark (2025) examples, this exercise has a medium scale VAR; 10 variables + constant
- Estimating the VAR at $p = 12$ we approach the feasible limit: only $T_u - k = 38$ degrees of freedom per equation, and a total number of coefficients $n + n^2 p = 1210$
- Bootstrap Confidence Intervals actually shrink!
- Confirming De Graeve and Westermark (2025) p.2 JME point on variance reduction even in a limit case "Misspecification reduction not only leads to Bias-Variance trade-off, but it reduces variance"

# Long Rung Identification Fragilities

# VAR IRFs; Neutral Technology Shock

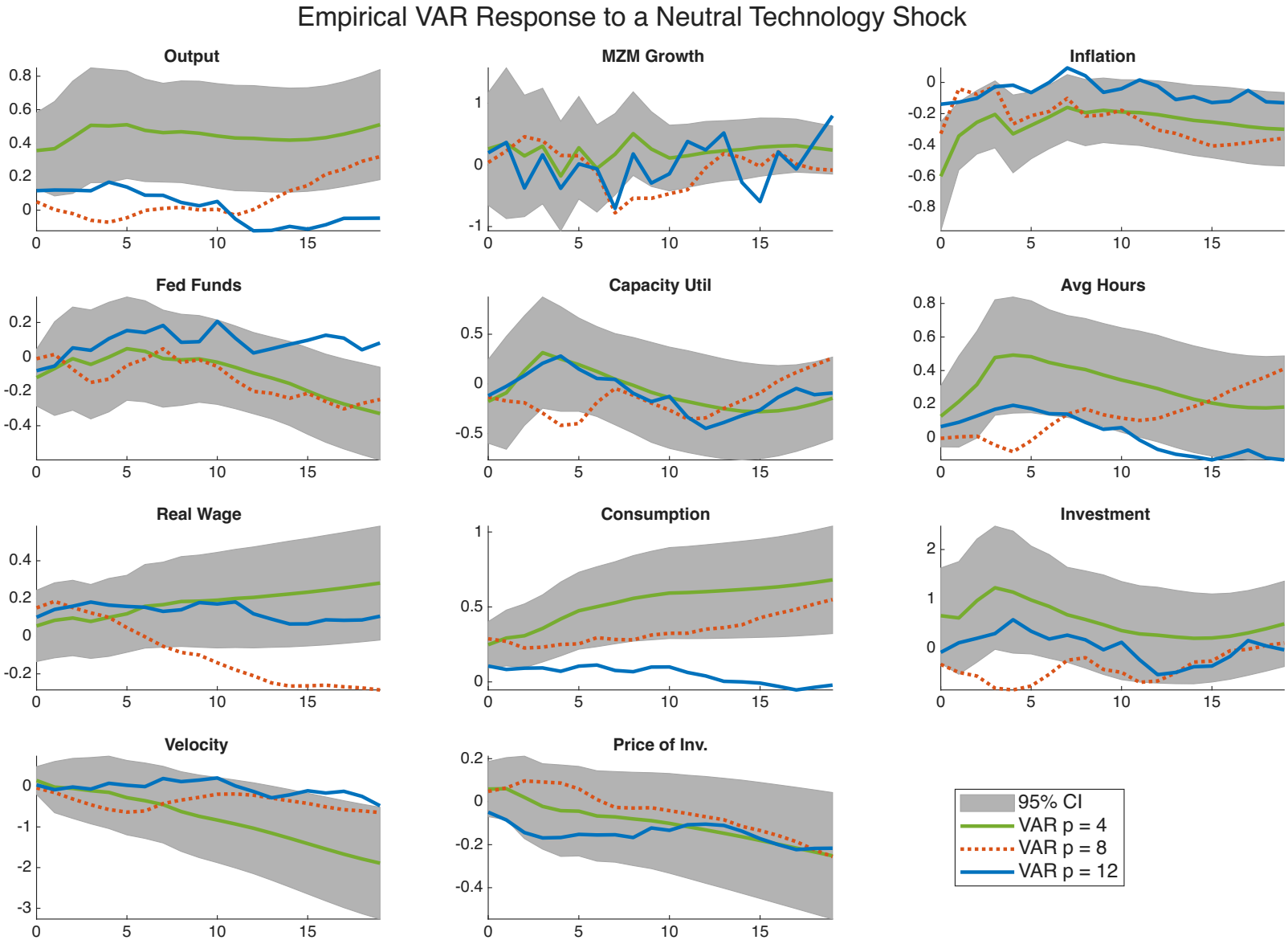



Figure 4: IRFs VARs all CI Comparison: Neutral Technology Shock

# VAR IRFs; Embodied Technology Shock

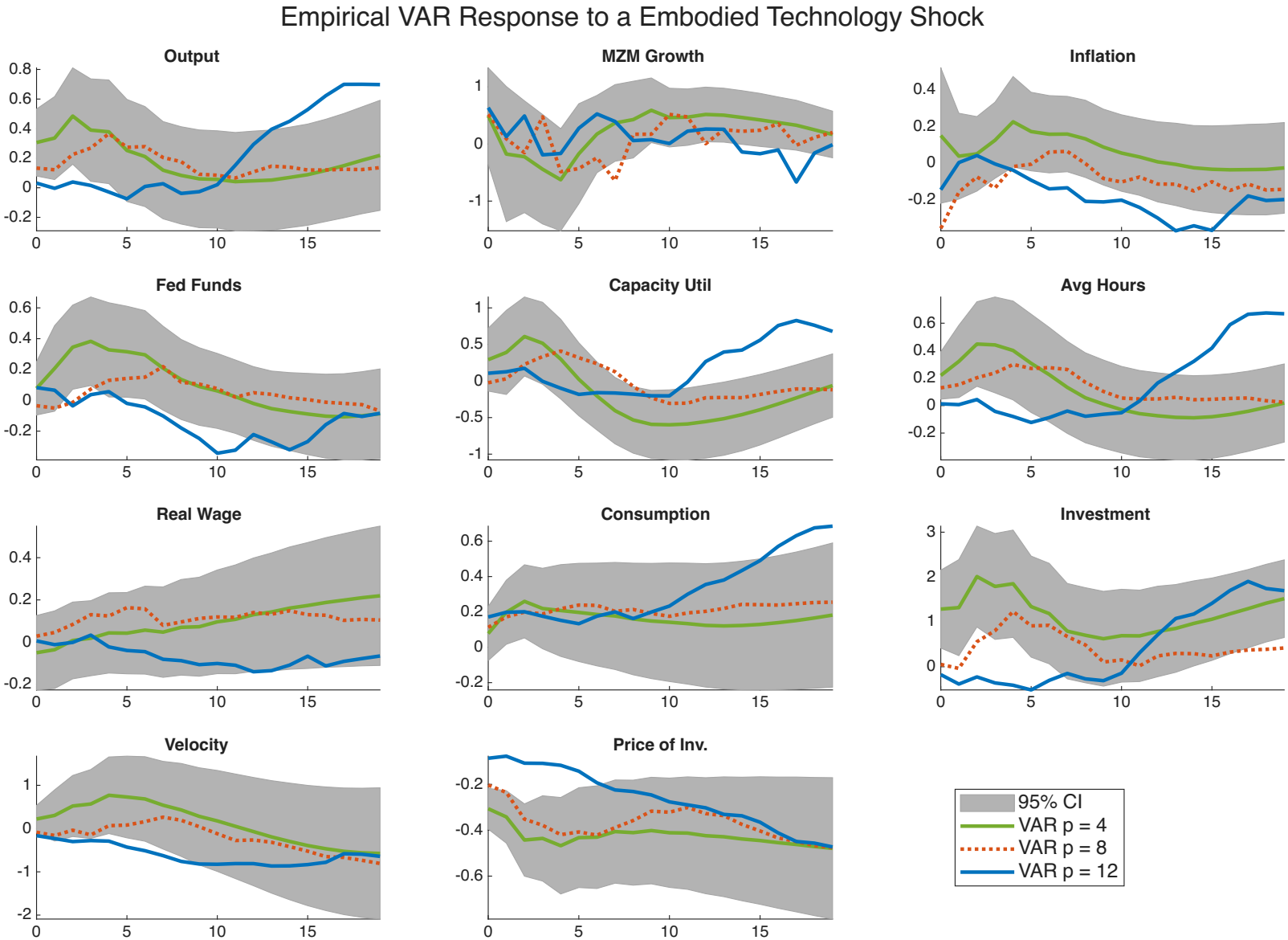



Figure 5: IRFs VARs all CI Comparison: Embodied Technology Shock

# DSGE IRFs; Monetary Policy Shock

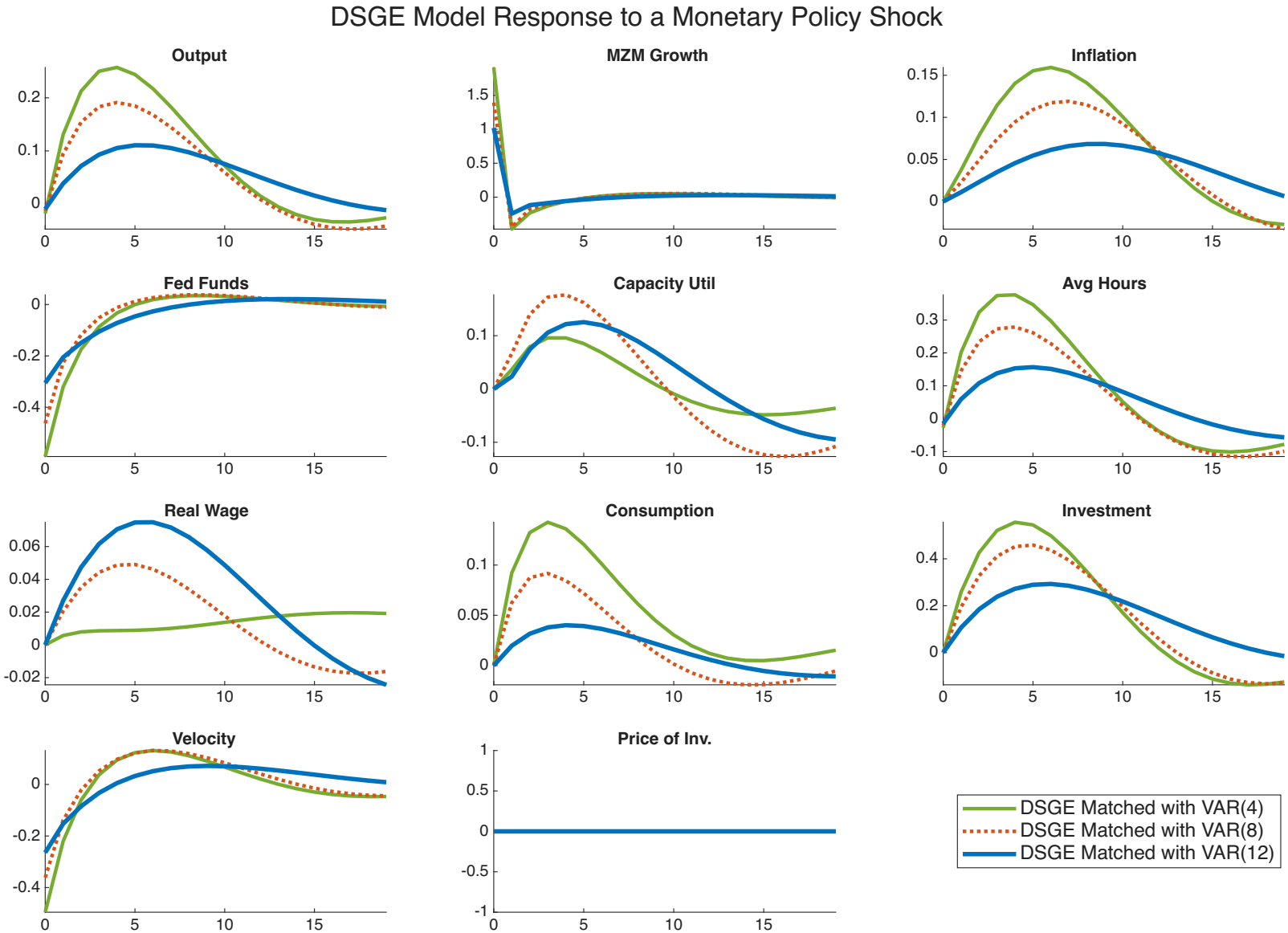



Figure 6: DSGE IRFs Long Lag Progressively Plots 4 8 12

# MonteCalvo

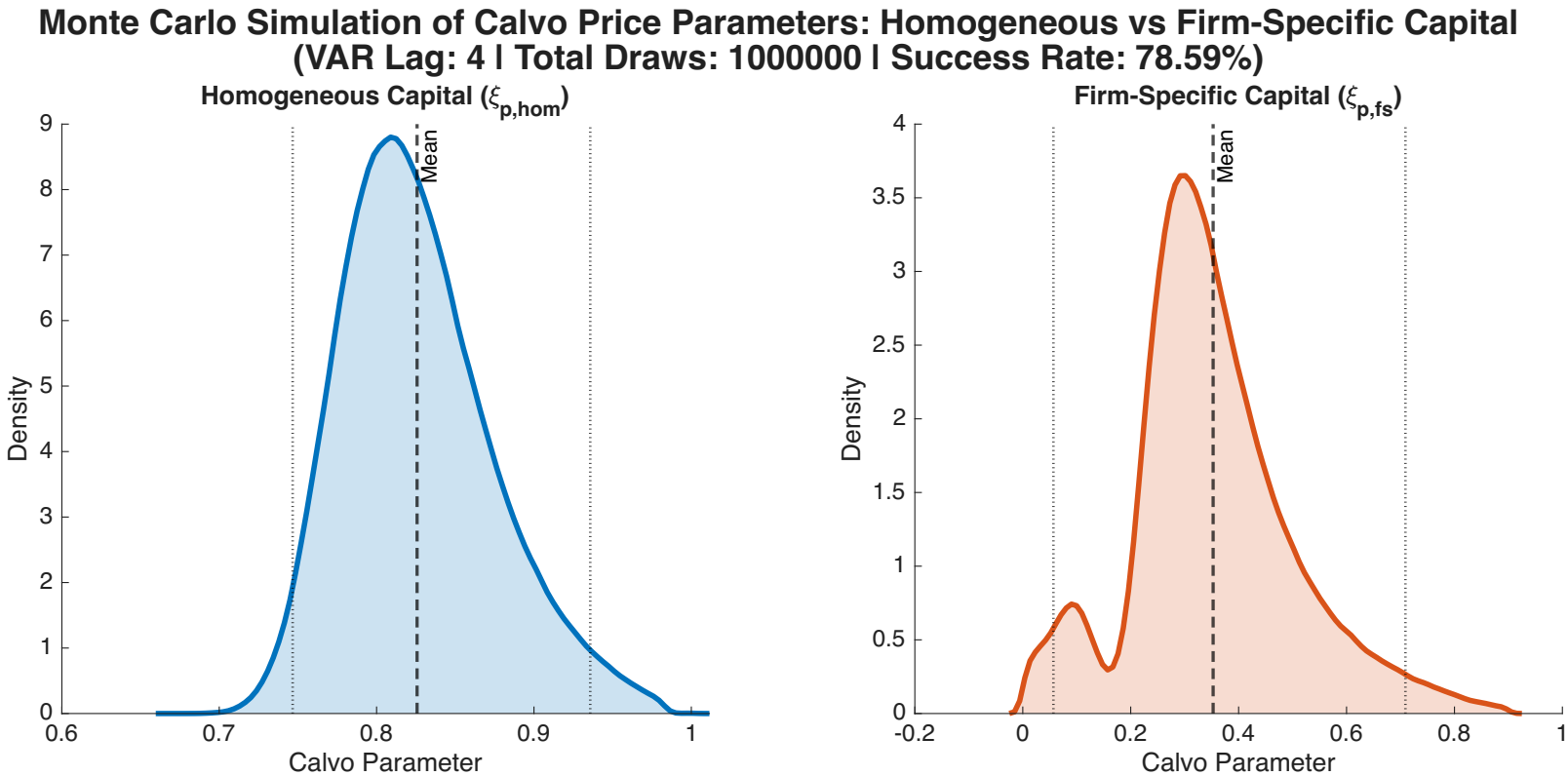



Figure 7: Calvo Price Parameter PDF from Monte Carlo Simulation; VAR(4)

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# MonteCalvo

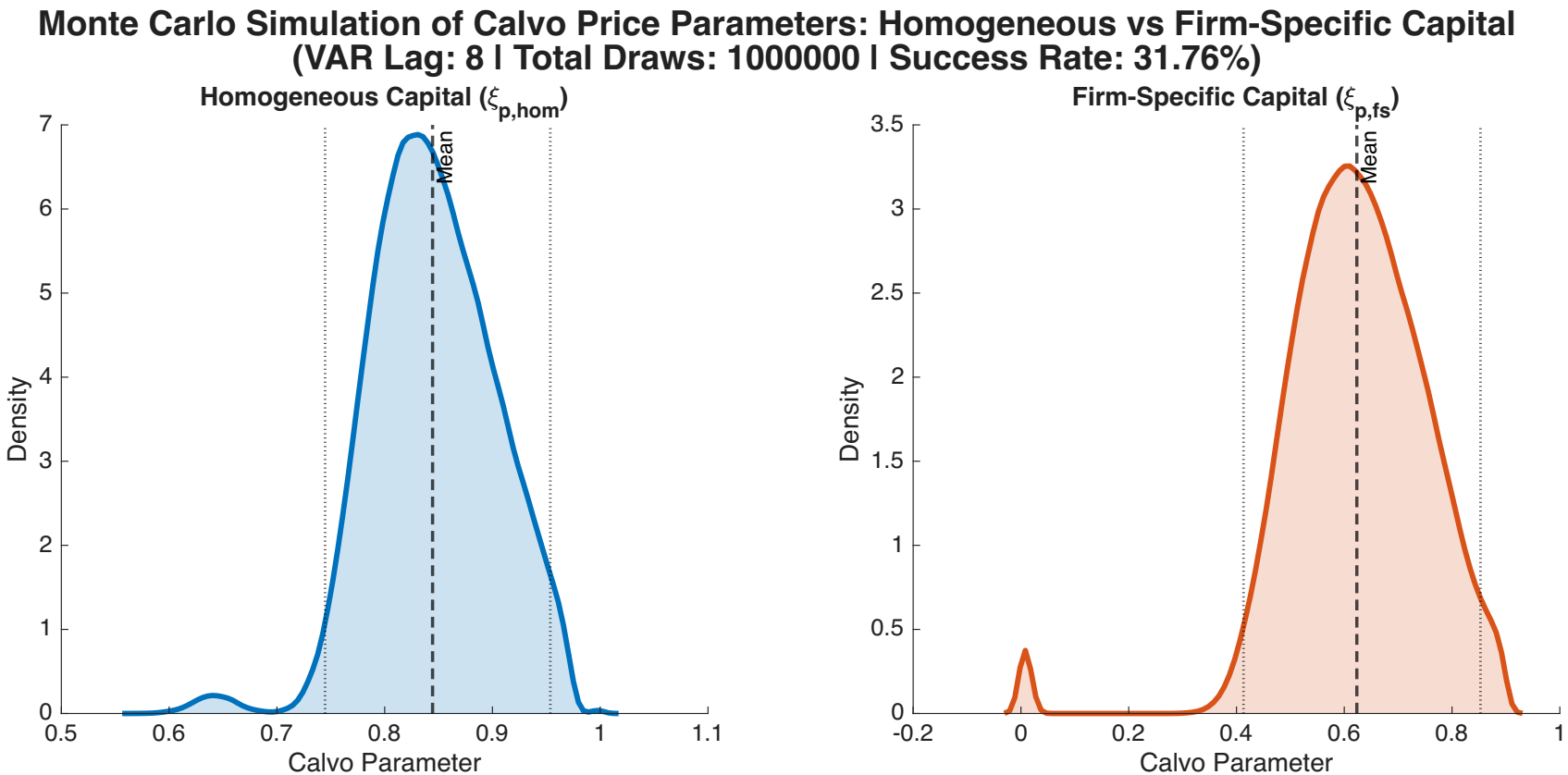



Figure 8: Calvo Price Parameter PDF from Monte Carlo Simulation; VAR(8)

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# MonteCalvo

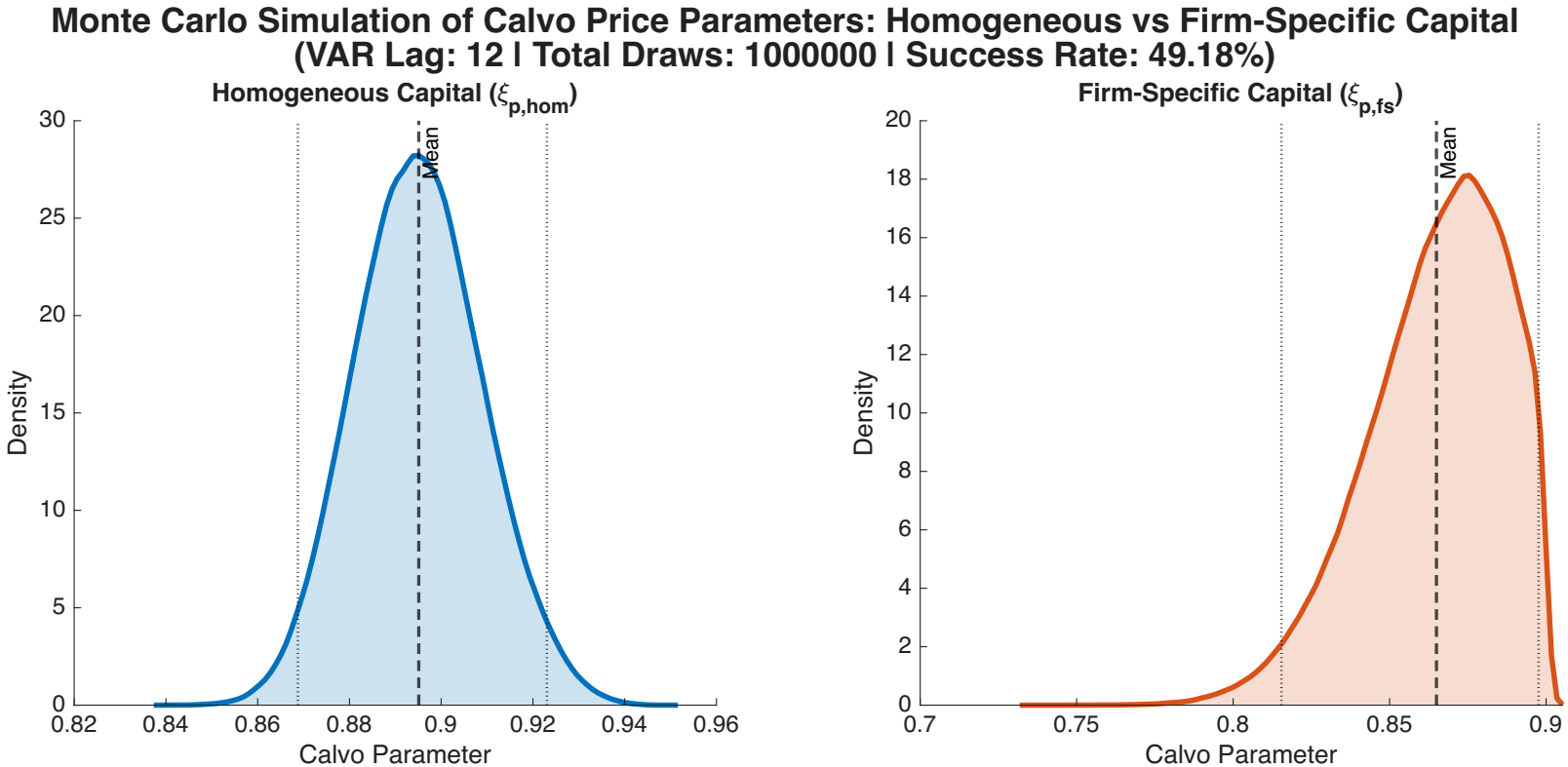



Figure 9: Calvo Price Parameter PDF from Monte Carlo Simulation; VAR(12)

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